Title page

Title: Outcome Bias in AI Models Applying a Culpability Decision Tree to Airline Occurrence Reports: A Pre-registered Within-Case Experiment

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Declarations

Declaration of competing interest. The author is the founder of CitadelAero, which develops AI tools for aviation safety and compliance teams. The study was carried out as a CitadelAero study, under the company's research procedures, and the company cites its findings in its own published material on the risks of general-purpose AI in aviation safety work. No CitadelAero product was tested. The results were to be published whichever way they came out, and the claims are limited to the models and conditions tested.

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CRediT authorship contribution statement. James Le Cuirot: Conceptualization, Methodology, Investigation (data export), Data curation, Validation (checking every case against its source report), Software and Formal analysis (carried out by Claude models under the author's direction, as declared in the manuscript), Writing – original draft (drafted by Claude under the author's direction, as declared in the manuscript), Writing – review & editing.

Pre-registration and preprint. The study was registered on OSF on 1 October 2026 (`https://osf.io/s8q24`). The materials, data and code are on OSF (`https://osf.io/dhf38`, DOI 10.17605/OSF.IO/DHF38). An earlier version of the manuscript was posted as a preprint (version 1, 2 October 2026) in the same OSF project, under the title "Same Actions, Different Endings: Outcome Severity Shifts How AI Models Apply Reason's Culpability Decision Tree to Airline Flight Crews". A plain-language write-up of the results is on the CitadelAero website.

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